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WellCheck: Smart Health Monitoring Management System (Cross-Platform)

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Abstract — This paper presents WellCheck, a cross-platform smart health monitoring management system developed to improve healthcare accessibility and patient care delivery in Malaysia. Existing healthcare platforms such as MySejahtera and MyHealth360 are limited by platform inconsistency, lack of real-time health monitoring, and absence of sensor integration, leaving a significant gap in continuous and accessible healthcare management. WellCheck addresses these limitations through a dual-platform architecture comprising a web application built with Spring Boot and Thymeleaf following the Model-View-Controller (MVC) pattern, and a mobile application developed using Flutter following the Model-View-ViewModel (MVVM) pattern, both sharing a centralized MongoDB Atlas database. The system integrates Arduino-based sensor hardware to automatically capture real-time vital health parameters including body temperature, heart rate, oxygen saturation, and ECG readings. An AI-powered disease prediction API developed using Django and Python provides additional diagnostic support. The system was developed using the Dynamic Systems Development Method (DSDM) across four iterative sprints, supporting five user roles: Patient, Doctor, Pharmacist, Clinic Assistant, and Administrator. Black-box testing validated all functional modules across five test packages, and User Acceptance Testing comprising 42 test cases was conducted with a clinic stakeholder, confirming system reliability and usability. WellCheck demonstrates that a cross-platform approach combining sensor technology, cloud database synchronization, and AI-assisted prediction can effectively bridge the healthcare accessibility gap for both urban and rural communities.

Keywords — Smart Health Monitoring, Cross-Platform, Flutter, MVVM, Spring Boot, MongoDB, Disease Prediction, Appointment Management, Sensor Integration, Role-Based Access.

I. INTRODUCTION

Healthcare accessibility remains a critical challenge globally, with digital health technologies increasingly recognized as a viable solution to bridge the gap between patients and healthcare providers [1]. In Malaysia, this challenge is particularly evident in the disparity between rural and urban healthcare access. According to the Ministry of Health Malaysia, rural residents require an average of 43 minutes to reach a hospital compared to 28 minutes for urban residents, highlighting an urgent need for continuous and remote healthcare support [2]. This gap is further compounded by the growing prevalence of chronic diseases that require routine monitoring, where infrequent hospital visits may result in missed critical changes in a patient's health condition.

The digital transformation of healthcare has introduced new opportunities to address these challenges through web-based and mobile platforms that enable remote patient monitoring, streamlined clinical workflows, and improved communication between patients and healthcare providers [3]. Mobile health technologies in particular have demonstrated significant effectiveness in improving healthcare service delivery and patient outcomes [11]. However, existing platforms in Malaysia such as MySejahtera operate exclusively as mobile applications, restricting access for desktop users and limiting the continuity of care across devices [4]. Furthermore, most existing systems lack real-time automated health parameter monitoring through sensor integration, leaving healthcare providers without timely data to make informed clinical decisions [12].

IoT based health monitoring systems have shown considerable promise in enabling real-time capture of vital health parameters such as heart rate, body temperature, oxygen saturation, and ECG readings for remote patients [5]. When combined with machine learning based disease prediction, such systems can further support early diagnosis and proactive health management [6]. Despite these advances, few existing healthcare platforms integrate sensor technology, AI based prediction, and cross platform accessibility into a single unified system.

This paper presents WellCheck, a cross platform smart health monitoring management system that addresses these limitations through three key contributions: (1) a dual platform architecture comprising a web application built with Spring Boot [7] following the Model View Controller (MVC) pattern and a mobile application developed using Flutter [8] following the Model View ViewModel (MVVM) pattern, both synchronized through a shared MongoDB Atlas cloud database [9], (2) integration of Arduino based sensor hardware for automatic real time capture of vital health parameters, and (3) an AI powered disease prediction API to support clinical decision making. The system was developed using the Dynamic Systems Development Method (DSDM), an agile framework that supports iterative and incremental delivery across complex multi-platform projects [13]. WellCheck supports five user roles namely Patient, Doctor, Pharmacist, Clinic Assistant, and Administrator, and integrates Stripe for secure payment processing [10], ensuring a comprehensive and accessible healthcare management experience across both web and mobile platforms.

II. PROBLEM BACKGROUND

Healthcare facilities in Malaysia continue to rely heavily on manual record keeping practices, which introduce errors, reduce operational efficiency, and limit the continuity of patient care [2]. Patients with chronic conditions who require routine monitoring are particularly affected, as the absence of a centralized digital system means that health data is fragmented across multiple visits and locations, making it difficult for healthcare providers to maintain accurate and up to date patient records [3].

Real time health monitoring remains largely absent in most existing healthcare systems. Without sensor technology, critical changes in a patient's vital signs such

as fluctuations in heart rate, body temperature, or oxygen saturation may go undetected between medical appointments, posing significant risks for high-risk patients [5]. The reliance on manual observation during scheduled visits alone is insufficient for effective chronic disease management and timely clinical intervention.

Communication between patients and healthcare providers is also hindered by the lack of a unified platform. Existing systems such as MySejahtera operate solely on mobile devices, preventing patients who rely on desktop access from managing their health records or appointments [4]. This platform fragmentation results in inconsistent user experiences and limits the ability of healthcare providers to access patient data across different devices and environments.

Furthermore, the absence of AI assisted diagnostic support in current platforms means that disease prediction and early detection still depend entirely on manual clinical assessment. This reduces the potential for proactive health management and places additional burden on healthcare professionals, particularly in settings with high patient volumes.

III. PROJECT AIM

The aim of this project is to develop WellCheck, a cross platform smart health monitoring management system that integrates web and mobile applications with real time sensor technology to improve patient care, streamline healthcare administration, and enhance communication between patients and healthcare providers across different devices and platforms.

IV. PROJECT OBJECTIVE

The objectives of this project are:

- i. To investigate existing healthcare systems and identify requirements for streamlining administrative processes by digitizing appointment management, prescriptions, and patient records to reduce manual work and improve overall efficiency.
- ii. To design and develop a cross-platform health monitoring system that integrates sensor technology for patients, doctors, pharmacists, and administrators.
- iii. To test and validate the system functionality to ensure it meets requirements and operates reliably across different platforms.

V. PROJECT IMPORTANCE

WellCheck serves patients seeking continuous health monitoring, healthcare providers requiring real time access to patient data, and clinic administrators managing appointments and records across web and mobile platforms. The system eliminates the need for repeated hospital visits by enabling patients to monitor their vital health parameters remotely through integrated Arduino based sensor hardware that automatically captures body temperature, heart rate, oxygen saturation, and ECG readings in real time.

The integration of an AI powered disease prediction API provides additional diagnostic support, allowing healthcare providers to make more informed clinical decisions based on analyzed patient data. This capability is particularly significant for patients in rural areas who face limited access to specialist consultations and require proactive health management between scheduled visits.

WellCheck also improves operational efficiency by replacing manual record keeping with a centralized digital platform that synchronizes data across web and mobile interfaces through a shared MongoDB Atlas database. Appointment management, prescription handling, patient records, and payment processing via Stripe are all consolidated within a single system accessible to five user roles namely Patient, Doctor, Pharmacist, Clinic Assistant, and Administrator. This unified approach ensures consistent functionality and user experience across devices, addressing the platform fragmentation observed in existing healthcare systems in Malaysia.

VI. LITERATURE REVIEW

Several existing healthcare platforms were analyzed to identify gaps and inform the development of WellCheck.

MySejahtera is the largest healthcare application in Malaysia, offering features such as health status tracking, appointment booking, and COVID-19 contact tracing [4]. However, MySejahtera operates exclusively as a mobile application, preventing desktop users from accessing healthcare services through a web browser. This platform restriction limits accessibility for users who rely on computers for their daily activities.

MyHealth360 is a Malaysian healthcare platform that provides both web and mobile interfaces for appointment management and medical record access.

Despite offering dual platform support, MyHealth360 suffers from inconsistent features across its web and mobile versions, resulting in a fragmented user experience where certain functionalities are only available on specific platforms.

IoT based health monitoring systems have demonstrated significant potential in enabling real time capture of vital health parameters for remote patient monitoring [5]. The integration of wearable and embedded sensors with cloud-based platforms allows continuous health data collection and transmission, supporting timely clinical intervention and chronic disease management [12].

Machine learning based disease prediction has also been widely explored as a means of supporting early diagnosis and reducing dependence on manual clinical assessment [6]. When integrated into a health monitoring system, AI powered prediction models can analyse patient data and provide diagnostic insights that improve the overall quality of healthcare delivery [1].

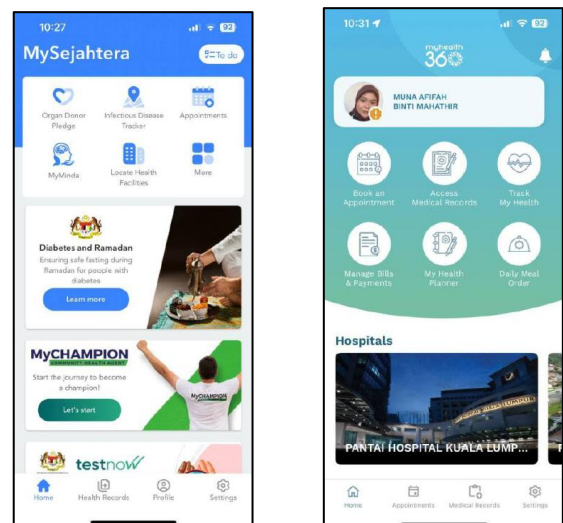


Fig. 1. MySejahtera and MyHealth360 Application Interface

VII. CURRENT SYSTEM ANALYSIS

Based on the analysis of the two existing systems, MySejahtera and MyHealth360 were identified as the most comparable platforms to WellCheck due to their focus on healthcare management in the Malaysian context. However, both platforms share critical limitations. MySejahtera lacks web browser support, restricting access to mobile users only, while MyHealth360 provides inconsistent features across its web and mobile versions resulting in a fragmented user experience. Neither platform supports real time

automated health parameter monitoring through sensor integration, AI assisted disease prediction, or a unified cross platform experience. WellCheck addresses all of these gaps by combining sensor technology, AI based prediction, and a synchronized dual platform architecture into a single comprehensive healthcare management system.

VIII. COMPARISON BETWEEN SYSTEM

Table I presents a feature comparison between the existing systems and WellCheck.

Table I. Comparison Between Existing Systems and WellCheck

Feature/System	MySejahtera	MyHealth360	WellCheck
User Registration	Yes	Yes	Yes
User Profile Update	Yes	Yes	Yes
Daily Symptoms Tracking	Yes	No	Yes
Real Time Status Health Parameter	No	No	Yes
Diagnose Ailments and Prescription Platform	No	Yes	Yes
Appointment Management	Yes	Yes	Yes
Email Notifications	Yes	Yes	Yes
Billing System	No	Yes	Yes

IX. METHODOLOGY

The Dynamic Systems Development Method (DSDM) was selected as the development methodology for the WellCheck Smart Health Monitoring Management System. DSDM is an agile framework that organizes development into fixed length iterative sprints and emphasizes continuous stakeholder involvement throughout the project lifecycle [13]. It was chosen because the nature of this project required flexibility to accommodate evolving requirements across two platforms simultaneously, involving multiple user roles including patients, doctors, pharmacists, clinic assistants, and administrators. By prioritizing early delivery of functional software, DSDM enabled the development team to progressively build and validate system features, collect meaningful feedback, and refine the system without causing delays to the overall project schedule.

The iterative structure also aligned well with the academic semester timeline of PSM 2, allowing systematic progress tracking across four development sprints covering user management, health monitoring, appointment management, and financial management modules.

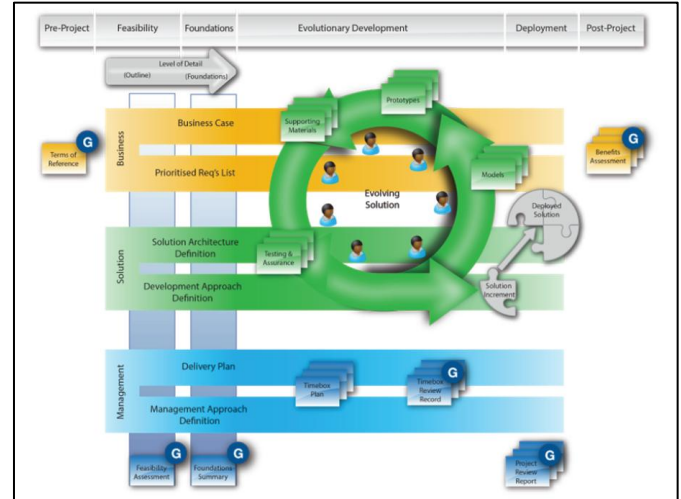


Fig. 2. DSDM Development Framework

X. TECHNOLOGY USED

Table II summarizes the main technologies used to build WellCheck, including the frameworks for the web and mobile applications, the database, the AI prediction backend, and the tools used for development and project management.

Table II. Technology Used in Wellcheck

Technology	Description
Spring Boot	Java based framework for developing the web application backend following the MVC architectural pattern.
Thymeleaf	Server-side Java template engine for rendering dynamic web pages as the View layer in the web application.
Flutter	Cross platform mobile development framework for building the mobile application following the MVVM architectural pattern.
Node.js with Express	ExpressServer side JavaScript runtime for building the mobile backend RESTful API.
MongoDB Atlas	Cloud based NoSQL database for centralized data storage and synchronization across web and mobile platforms.
Arduino	Microcontroller hardware for integrating ECG, body temperature, heart rate, and oxygen saturation sensors for real time health data capture.
Django	Python based framework for developing the AI disease prediction API backend.
Stripe	Payment gateway for secure online billing and payment processing within the web and mobile applications.

Visual Studio Code	Primary integrated development environment for both web and mobile application development.
GitHub	Version control platform for source code management and team collaboration.
Enterprise Architect	UML modelling tool for designing system architecture diagrams and documentation.
JIRA	Project management and issue tracking tool used throughout the development lifecycle.

XI. REQUIREMENT ANALYSIS

A. Functional Requirements

Table III outlines the functional requirements of the WellCheck system.

Table III. Functional Requirements of WellCheck

ID	Description
FR001	The system shall allow users to register and login across web and mobile platforms.
FR002	The system shall allow patients to submit daily health symptoms.
FR003	The system shall allow patients to view doctor prescriptions and diagnosis.
FR004	The system shall allow patients to book and manage appointments.
FR005	The system shall allow patients to monitor real time health parameters via sensor integration.
FR006	The system shall allow patients to make payments via Stripe.
FR007	The system shall allow doctors to assign patients and send prescriptions.
FR008	The system shall allow pharmacists to manage medicine stock.
FR009	The system shall allow clinic assistants to book appointments on behalf of patients.
FR0010	The system shall allow administrators to manage user accounts.
FR0011	The system shall send email notifications for appointment updates and prescription delivery.
FR0012	The system shall support AI based disease prediction via the Django prediction API.

B. Non-Functional Requirements

Table V outlines the non-functional requirements of the Wellcheck system.

Table V. Non-Functional Requirements of Wellcheck

ID	Description
NF001: Reliability	The system must remain operational without unexpected crashes during normal use.
NF002: Usability	The system shall be intuitive and accessible for users across all roles without technical background.
NF003: Portability	The system shall be accessible via any modern web browser and mobile device.
NF004: Availability	The system shall be accessible 24 hours per day, 7 days per week via cloud deployment.
NF005: Security	The system shall enforce role-based access control with secure authentication across platforms.

NF006: Performance	Real time sensor data shall be captured and synchronized across platforms without significant delay.
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C. Design

The WellCheck system adopts a dual platform architecture to support both web and mobile users. The web application follows the Model View Controller (MVC) pattern built with Spring Boot and Thymeleaf, while the mobile application follows the Model View ViewModel (MVVM) pattern built with Flutter. Both platforms share a centralized MongoDB Atlas database ensuring real time data synchronization across all user roles. The system supports five primary user roles namely Patient, Doctor, Pharmacist, Clinic Assistant, and Administrator, with role specific interfaces designed for each platform.

Fig. 3 shows the use case diagram of the WellCheck mobile application, illustrating how the Patient and Admin actors interact with the Health Monitoring, User Management, Appointment Management, and Financial Management modules, including sending daily symptoms, viewing health status, managing appointments, and making payments.

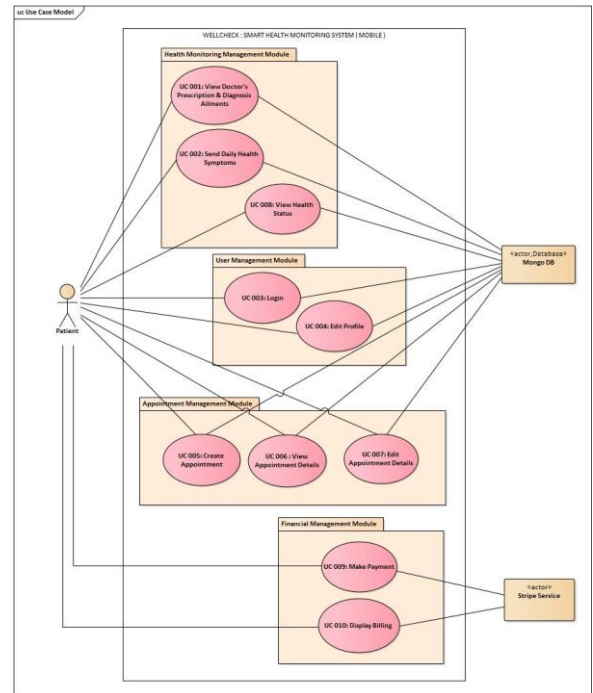


Fig. 3. Use Case Diagram of WellCheck (Mobile)

Fig. 4 shows the use case diagram of the WellCheck web application, illustrating the interactions between the Patient, Doctor, Pharmacist, Clinic Assistant, and Administrator actors with functions such as managing prescriptions, tracking symptoms, handling

Fig. 9 shows the Doctor Prescription page on the web application, where patients can view their diagnosis, prescribed medicine, and dosage instructions issued by their doctor.

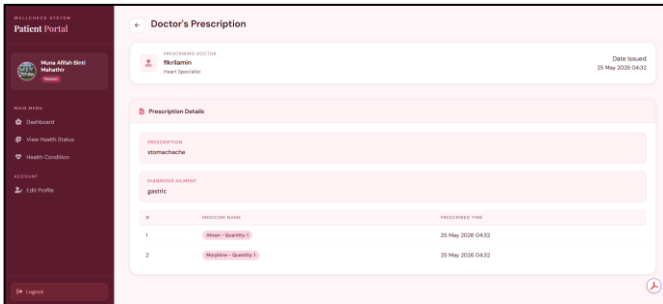


Fig. 9. View Doctor Prescription and Diagnosis (Web)

Fig. 10 shows the Health Condition page on the web application, displaying the patient's latest sensor readings, including temperature, oxygen saturation, and heart rate, along with a trend graph of recent health data.



Fig. 10. Health Condition Page (Web)

Fig. 11 shows key interfaces of the WellCheck mobile application, including the patient dashboard, the appointment booking screen, and the payment screen for pending appointment fees.

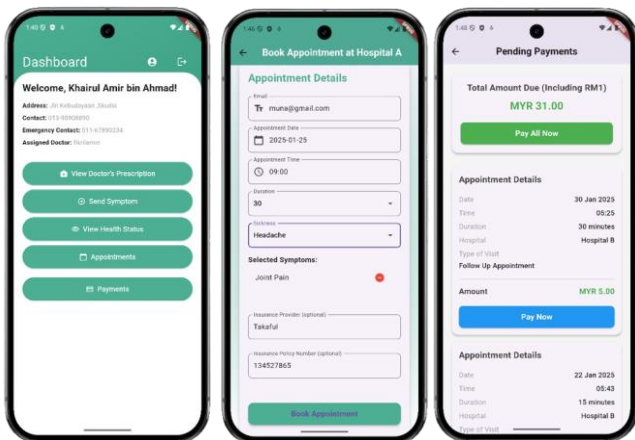


Fig. 11. Key Mobile Application Interfaces of WellCheck

XII. TESTING

System testing is conducted to validate the functionalities and features of the WellCheck Smart Health Monitoring Management System to ensure that the system fulfilled the end-to-end system specification with the agreement between the developers and stakeholders.

A. Black Box Testing

Black box testing is a software testing approach for testing and validating the system features and functionalities without referring to the code structure of the system. The testing was conducted across five major packages namely User Management, Health Monitoring Management, Appointment Management, Financial Management and Sensor Management.

Table VI. Black-Box Testing for Wellcheck

TC ID	Use Case	Result
TC001	Login - User Management Module	Pass
TC002	Edit Profile - User Management Module	Pass
TC003	Receive Email Notification - User Management Module	Pass
TC004	View Doctor Prescription and Diagnosis - Health Monitoring Module	Pass
TC005	Send Daily Health Symptoms - Health Monitoring Module	Pass
TC006	View Health Status - Health Monitoring Module	Pass
TC007	Create Appointment - Appointment Management Module	Pass
TC008	View Appointment Details - Appointment Management Module	Pass
TC009	Edit Appointment Details - Appointment Management Module	Pass
TC0010	Update Appointment - Appointment Management Module	Pass
TC0011	Make Payment - Financial Management Module	Pass
TC0012	Display Billing - Financial Management Module	Pass
TC0013	Set Sensor - Sensor Management Module	Pass
TC0014	Register Sensor - Sensor Management Module	Pass
TC0015	View Sensor Status - Sensor Management Module	Pass
TC0016	Accept Sensor Agreement - Sensor Management Module	Pass
TC0017	Book Appointment - Appointment Management Module	Pass

B. User Acceptance Testing

User Acceptance Testing (UAT) was conducted with a clinic stakeholder as the real-life end user of the system. All five user roles namely Patient, Doctor, Pharmacist, Clinic Assistant, and Administrator were tested. A total of 42 test cases were evaluated across all five system modules to confirm that the system meets the functional requirements and usability expectations of real-world healthcare operations. All test cases returned a Pass result, confirming that WellCheck operates reliably and meets user needs across both web and mobile platforms.

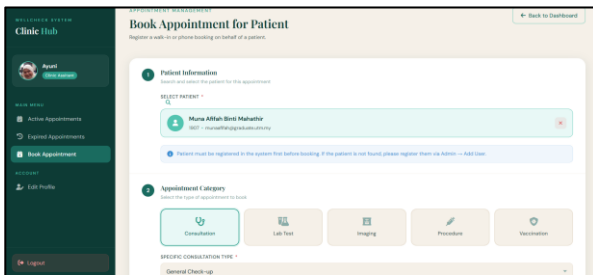


Fig. 12. System Interface of Book Appointment by Clinic Assistant

2.7.3 TC007_03: Successfully Create Appointment (Clinic Assistant)			
Test Case ID	TC007_03	Test Case Description	Verify Clinic Assistant can successfully book an appointment on behalf of a patient through the WellCheck web interface.
Created by:	Muna Afifah Binti Mahathir	Version:	2.0
Tested by:	Noor Azmina Ezzaty	Date:	16.06.2026
No	Pre-requisites	No.	Test Data
1.	Clinic Assistant is logged into WellCheck web interface.	1.	Patient Email: muna@gmail.com.
2.	Patient account exists in the system.	2.	Appointment Date: 2025-01-25.
3.	Clinic Assistant has appointment management permissions.	3.	Appointment Time: 09:00.
4.	MongoDB server is operational.	4.	Duration: 30 minutes.
5.	Network connection available.	5.	Appointment: X-Ray
		6.	Insurance Provider: Takaful
		7.	Insurance Policy Number: 134527865
Test Conditions	1. Clinic Assistant selects a patient from the system to book an appointment on their behalf. 2. Clinic Assistant fills in all required appointment details through the web form. 3. Appointment is visible to both the patient and the assigned doctor after booking.		
Step #	Step Details	Expected Result	Actual Result
1.	Clinic Assistant logs into WellCheck web dashboard.	Clinic Assistant dashboard displayed successfully.	Clinic Assistant dashboard displayed successfully.
2.	Clinic Assistant navigates to Appointment Management section via sidebar.	Appointment management page loaded showing all existing appointments.	Appointment management page loaded showing all existing appointments.

Fig. 13. STD Test Case for Book Appointment by Clinic Assistant

No	Actions	Expected Results	Pass / Fail	Comment
20.1	Update Appointment Status 1. Clinic Assistant logs into WellCheck web dashboard. 2. Clinic Assistant navigates to Active Appointments via sidebar. 3. Clinic Assistant locates the appointment for patient007 with appointment date 2026-06-05 at 09:30. 4. Clinic Assistant clicks the tick (✓) button under Actions column.	1. Clinic Assistant dashboard displayed successfully as shown in Figure 3.0 (a). 2. Appointment management page loaded showing all existing appointments as shown in Figure 3.0 (b). 3. Appointment record displayed in the list with current status pending approval.	Pass	

Fig. 14. UAT Test Case for Update Appointment Status by Clinic Assistant

XIII. CONCLUSION

This paper presented WellCheck, a cross platform smart health monitoring management system that addresses the limitations of existing healthcare platforms in Malaysia. The system integrates a Spring Boot web application following the MVC pattern and a Flutter mobile application following the MVVM pattern, both synchronized through a shared MongoDB Atlas database. Arduino based sensor hardware enables real time capture of vital health parameters including body temperature, heart rate, oxygen saturation, and ECG readings, while an AI powered disease prediction API provides additional diagnostic support for healthcare providers.

The system was developed using the DSDM methodology across four iterative sprints and validated through black box testing covering 16 test cases across five modules and User Acceptance Testing comprising 42 test cases conducted with a clinic stakeholder. All test cases returned a Pass result, confirming that WellCheck meets both technical requirements and real-world usability expectations across all five user roles.

Future work includes the addition of multilingual support, expansion of the AI prediction module to support a wider range of disease classifications, and the development of dynamic health trend graphs to provide patients and doctors with deeper insights into long term health patterns. WellCheck demonstrates that a cross-platform approach combining sensor technology, cloud database synchronization, and AI assisted prediction can effectively bridge the healthcare accessibility gap for both urban and rural communities in Malaysia.

XIV. ACKNOWLEDGEMENT

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